



SAT Digital Course Pack

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Studyworks Prep

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About the Digital SAT

The Digital SAT is a computer-based test that measures student abilities in Math and Reading/Writing. The test emphasizes problem-solving and close reading ability, rather than strict academic knowledge. Here is a basic breakdown of the Digital SAT:

- The test is scored on a 1600-point scale, with two subscores (Reading/Writing and Math) on an 800-point scale.
- Students take the test on their own device (laptop or tablet), or one provided by the College Board. I recommend you bring a device with a physical keyboard and a larger screen, to give you more room to work on questions.
- The test is shorter than any previous version. Here's a rundown of the format:

Section	Length	Number of Questions
Reading and Writing	64 minutes (32 mins per module)	54
Math	70 minutes (35 mins per module)	44
Total	134 minutes, plus 3 short breaks	98

- The test is **adaptive**, which means that the questions change depending on your performance. Each section of the SAT (Reading/Writing and Math) is divided into two Modules. The first Module of each section will be identical for all students and will include a mixture of Easy, Medium, and Hard problems. Depending on how a student performs on Module 1, the second Module will adapt to provide more questions at the student's skill level. For example, if a student gets most of the questions in Module 1 correct, the test will provide more Hard-level questions in the second module. The higher difficult questions are worth more points than lower difficulty ones, so it's important to do well in the first module and unlock the more valuable content.
- Online practice materials are available to all students free of charge. Visit satsuite.collegeboard.org to find online tests and other materials. You will need the Bluebook app (found in the App Store for your individual device) to do the practice tests.
- A calculator is allowed for all math questions. Though students can bring their own calculators, the testing application includes a graphic calculator powered by Desmos, which I encourage all students to use and master. As of this writing, I can think of no situation where a physical calculator would be preferable to Desmos.

Essential Skills for Reading and Writing

The Reading and Writing section contains a mix of reading comprehension and English grammar/usage questions. The exact distribution of questions varies from test to test, but the general pattern has remained constant:

1. Sentence Completion
2. Reading Comprehension
3. Grammar/Usage
4. Research Notes

In general, the test favors students who are patient, careful readers with a good eye for detail. It's important to remember that most questions are easier and more literal than they appear, but students have a tendency to give up too early and settle for an attractive answer choice. Students also tend to skim or completely skip over details in the passage and the questions: their eyes keep moving, but none of the information is actually being processed. This happens especially toward the end of answer choices, which is why many wrong answer choices are designed to sound great at first, then fail at the very end. But don't despair! You can become a more disciplined and careful reader through practice, and the work you put into improving your reading discipline will help you for the rest of your life.

The Big Secret of Reading Comprehension Questions

Every Reading Comprehension question contains a single, unambiguous right answer that is directly supported by the text. Every part of the answer will be provably correct, based on the most literal possible reading of the text. Every wrong answer will have at least one false or unsupported detail, or it requires an assumption far beyond the literal information in the text. Don't be creative!

The Big Secret of Grammar/Style/Punctuation Questions

In questions that directly ask about the essential rules of English writing, the right answer will follow a **very specific** set of rules that the SAT has chosen. These rules don't necessarily correspond to the way people normally write, even in pretty formal settings like newspaper articles and textbooks. Instead, the rules represent the SAT's unique interpretation of what "good writing" is, and therefore they have to be memorized. When facing these types of questions, it is crucial to apply the rules **exactly**, with no wiggle room whatsoever.

Reading Writing Skill 1: Words in Context

Every Reading/Writing module begins with between three and eight “Words in Context” questions. Most of these are the classic “Sentence Completion” type, where you’re given a sentence with a missing word or phrase, and you have to complete the text. A small number of “Words in Context” questions are the “Most Nearly Means” type, where you’re given a sentence with an underlined word or phrase, and you have to choose an alternative phrasing that has the same meaning.

Although these questions may seem very different from typical reading questions, they’re actually quite similar in the way we approach them. Try these tips:

1. **Read the entire passage carefully, summarizing each sentence as you go.**

The answer to sentence completion questions often depends on the meaning of the passage as a whole. That is, you shouldn’t just skim the text, assuming the right answer will be an obvious synonym for a word in the passage. Use our summary strategies (See Skill 2) to develop a good understanding of the text as a whole.

2. **Look for keywords that provide clues about the missing or underlined word.**

Keywords come in different varieties:

- a. **Actors and viewpoints:** Look for the different players or actors in the sentence. It’s quite common to see two different viewpoints present, and the correct answer depends on keeping those viewpoints separate. Are there two groups of scientists? They probably have opposing views. Is there a specific book, then an essay about that book? The essay probably contains an analysis or opinion of the book. Try to keep the actors separate in your mind.
- b. **Logical connectors:** Some phrases tell you that there is a change or difference between elements (“on the other hand,” “however,” “other researchers...”). Other phrases tell you that there is a cause and effect relationship (“as a result,” “accordingly”) between the elements. Always underline or highlight these connectors, because they can change the meaning of the whole sentence.
- c. **Negators:** If there is a negating phrase right before the blank (“not,” “never”), the meaning of the blank is “flipped” to its opposite. Whatever the sentence is trying to say as a whole, the word in the blank is likely to be the exact opposite.
- d. **Synonyms and antonyms:** In many passages, there is an exact word or phrase that directly correlates to the blank. That word or phrase might be a simple synonym for

the word in the blank, or it might be the exact opposite of the word in the blank (usually, because there's a **negator** involved).

3. Using the clues as literally as possible, come up with your own word for the blank.

Your word should **fit logically and grammatically into the sentence**, and it should be as specific as possible to the clues you picked up. In other words, you should be able to read the sentence out loud with your own replacement, and it will sound perfectly natural and have the exact same meaning as the original sentence.

If you can't come up with a single word, you may still be able to think of a phrase that captures the meaning of the blank. For example, you might think that the word in the blank should mean something like "keeps you safe," and you'll see that one of the answer choices is "guards."

4. Read through the answer choices, looking for a word that matches your word.

Focus on the words you know first, eliminating anything that doesn't match your reading of the sentence. Next, see if any of the words you don't know contain roots and prefix/suffix parts that might allow you to make an educated guess at what they mean. If you have a "feeling" about what a word means, try not to use that unless you literally have nothing else.

5. Keep your eyes open for words that coincide with specific language in the passage.

Sometimes, we miss the very important and obvious clue in the sentences, so we might end up looking for the wrong word or idea. If you see an answer that actually matches a specific phrase or word in the passage, give it some extra attention. Don't get so focused on your own prediction that you blind yourself to alternative interpretations.

Lazy Person Hint: Don't have time or energy for all that? If you find that this section is using too much of your time, you can get about 75% of the accuracy by looking at only the **synonym / antonym** keywords and ignoring everything else. Note that this simpler strategy will just not work on the most difficult sentence completion questions, because those questions are written specifically to fool students into choosing simple synonyms.

Reading Writing Skill 2: Process and Pre-Answer all Reading Comprehension Questions

Reading comprehension questions typically begin somewhere between question 4 and question 8, just after the sentence completion questions. The keys to navigating these questions successfully are focused attention and systematic work. Many of the reading questions are very, very tedious or throw huge amounts of information at you. Don't let the test defeat you with boredom! If you're patient and you attack each question with a healthy aggression, you can get every single question right. Here's a rundown of the process:

1. Read the introduction to the passage, if present.

Before some passages, particularly poetry or fiction excerpts, there is a brief introduction that provides the author's name and a general bit of context for the passage. This information is often quite useful, and is usually not contained in the passage itself. Don't skip past it!

2. Read the question.

Most of the reading questions seem like some variation of "what does this passage mean?". However, there is actually quite a bit of variety in the specific tasks you have to accomplish. Take a moment to read the question carefully, so you know what to look for when you read the passage.

3. Read the passage, making sure you understand each sentence.

The Digital SAT features very short passages, no more than two or three sentences. However, it turns out that it's perfectly possible to get lost even in such a brief text. After you read each sentence, take a quick pause to make sure you understood what you just read. If you got very little information, go back and try again. It may be necessary to read the sentence in smaller parts, pausing to understand each part as you go. The effort you put into understanding the passage will certainly be worthwhile—first, because you'll have a much better chance of getting the question right, and second, because every time you put work into reading, you're becoming a better reader.

4. Determine **FOR YOURSELF** what you think the answer might be.

The SAT test writers are extremely talented at writing attractive wrong answers. They use proven strategies to generate answers that target specific misreadings, shortcuts, and general imprecisions that students are prone to. The best way to dodge most of these traps.... is to avoid reading the answers! At least, for a little while.

After reading the question, then reading the relevant text, make every effort to come up with YOUR OWN answer to the question. Be as specific as possible, and do NOT look at the answer choices to help you through this step. The idea is to conduct your thinking about the question without making yourself vulnerable to trap answers.

5. Find the answer choice that matches your predicted answer, eliminate any answers that don't match.

See if one of the answer choices fits the pattern or general meaning of your predicted answer. Along the way, make a careful judgment of each answer choice and eliminate any answers that just don't match the information in the text. Remember that you should not be making any inferences or connecting any dots. Be skeptical of the answers and ruthless in eliminating anything that seems contrary or unrelated to the text. Hint: See **“Good Cop / Bad Cop,”** covered in the next skill section.

You should be left with one clear, right answer that is unambiguously a match for the passage. It might even seem too obvious, to the point that you wonder how anyone could get the question wrong. Believe me: for each question, there are literally millions of students who get that question wrong. Don't be one of those students!

Awesome Practice Method: Want to get a little more practice with reading and summarizing short passages? Use the power of AI!

1. Find an SAT passage, either in a practice test or from the Educator Question Bank.
Highlight and **copy** the text of the passage (without the question and the answer choices).
2. Go to ChatGPT or your preferred humanlike digital being.
3. Enter this prompt:

I am going to provide you with an original passage and my summary of that passage. I want you to rate my summary on a scale of 1 to 100 for accuracy and completeness, and I would like you to explain what elements of the original passage I missed or misrepresented in my summary. Do not reduce my score on account of lack of specificity or omission of specific facts. Only the general structure and meaning of the passage needs to be captured.

4. If everything is going according to plan, ChatGPT should ask you for the original passage. **Paste** the SAT passage and submit it.
5. Again, if everything is hunky dory, ChatGPT will ask you for your summary. **Carefully and thoroughly** summarize the passage, sentence by sentence. You can replace specific phrases (“Biologist Jim Likeedabugs”) with an equivalent summary (“A scientist”) to shorten your word length.
6. See what ChatGPT thinks!

Our goal is to get to **90+ out of 100** every time. If you fell short of that number, read through the specific feedback and repeat the process with the same passage. If you managed to get a 90 or above, find another passage (maybe look for a harder one!) and repeat the process with the new passage.

This exercise will help you improve your attention to detail and reduce your tendency to “extrapolate” or make assumptions about the passages. As a final note: ChatGPT is very good at this kind of task, so take the feedback seriously. It’s tempting to say, “Well this stupid machine is stupidly telling me I did something wrong, when I’m actually perfect,” but I’ve found that ChatGPT is always right about these summaries. Don’t resent your failures—drink them up like protein shakes and get stronger!

Reading Writing Skill 3: “Good Cop, Bad Cop”

Once you have developed your own possible answer to a reading question, it's time to consider the answer choices one by one, using your predicted answer to narrow down the options and eliminate clearly unrelated material. As you do that, you have to maintain a generally skeptical attitude toward the answer choices, eliminating answers that contain even small wrong details.

In other words, you need to look for true and accurate information while simultaneously rejecting anything that rings false. We call this combination of perspectives **“Good Cop, Bad Cop.”**

The Rule: The correct answer to a reading question will be based on the information in the passage only, with no assumptions or unsupported statements. “Good Cop” focuses on what is correct, while “Bad Cop” ruthlessly eliminates anything that's not directly supported by the passage.

Strategy: Good Cop / Bad Cop

Start by reading the question carefully, then reading the passage. As you read, summarize longer sentences into shorter logical units (“sound bites”). When you've finished the passage, try to come up with **your own** answer to the question.

Next, read the answer choices, using Good Cop / Bad Cop:

Good Cop: Based on what you believe should be the right answer, check each answer choice to see how well it fits your prediction. Does it agree generally with what you were looking for? Mark or highlight answers that fit the prediction roughly.

Bad Cop: Aggressively eliminate answers that contain any wrong details. Look at each piece of the answer and make sure there is evidence for every single word. Don't ignore verbs or adjectives - everything has to be supported.

Reading Writing Skill 4: Recognize an Independent Clause

An independent clause, also known as a main clause, is a group of words that forms a complete thought and can stand alone as a sentence. It expresses a complete idea and contains a subject and a predicate. Here's a breakdown of the components of an independent clause:

Subject: The subject is the part of the sentence that indicates who or what the sentence is about. It typically includes a noun or pronoun.

Predicate: The predicate is the part of the sentence that provides information about the subject, including the action the subject is performing or a state of being. It usually includes a verb and may also include other elements like objects, complements, or modifiers.

Here are a few examples of independent clauses:

"She is studying for her exams."

Subject: "She"

Predicate: "is studying for her exams"

"The sun sets in the west."

Subject: "The sun"

Predicate: "sets in the west"

"The girl in the blue dress traveled to Europe last summer."

Subject: "The girl in the blue dress"

Predicate: "traveled to Europe last summer"

Each of these examples presents a complete thought and can function as a standalone sentence. It's important to note that an independent clause is distinct from a dependent clause, which does not express a complete thought and relies on an independent clause to form a complete sentence.

Note: The **imperative form** of a sentence, which is used to state a command, is considered an independent clause. For example:

"Eat my shorts."

- Bart Simpson

The subject of an imperative sentence is not stated, but is implied to be “you,” the listener or reader.

Reading Writing Skill 5: Identify the Noun Head in a Phrase

In grammatical terms, a "noun head" refers to the primary or central noun within a phrase or construction. Nouns are words that represent people, places, things, or ideas, and they often serve as the core or head of a noun phrase. A noun head is the main noun around which other words or elements gather to provide additional information or context.

Here are a few examples to illustrate the concept of a noun head in noun phrases:

Simple Noun Phrase:

"Dog"

In this case, "dog" is the noun head. The noun phrase consists only of the head itself without additional modifiers.

Noun Phrase with Modifiers:

"The big dog"

In this example, "dog" is still the noun head, but the adjective "big" modifies and provides more information about the head.

Noun Phrase with Determiner and Modifier:

"My friendly neighbor's white cat"

Here, "cat" is the noun head. The determiner "my," along with the adjectives "friendly," "neighbor's," and "white," modify or describe the head, providing more details.

Understanding the noun head is particularly relevant when analyzing the structure of noun phrases in syntax. The noun head typically carries the main meaning of the phrase, and other elements within the phrase, such as determiners, adjectives, and modifiers, only serve to enhance or specify that meaning. For the purposes of subject-verb agreement, only the noun head of a phrase is considered when determining the correct form of a verb.

Reading Writing Skill 6: Match a Subject to a Verb (Subject-Verb Agreement)

In any independent clause, the main verb has to agree in number (singular or plural) with its subject. The basic rule is that a singular subject should be paired with a singular verb, and a plural subject should be paired with a plural verb. This agreement helps maintain clarity and grammatical correctness in a sentence.

On the SAT, subject/verb agreement questions follow a very specific pattern that we can exploit for better accuracy and speed.

Subject/Verb Crusher:

1. Check whether all four answers are verbs. If they are, proceed to Step 2.
2. For each answer choice, check whether the verb is singular or plural. Use the “It/They” test to make sure.
3. If there are three answers of one type (plural or singular) and one answer of the other type, the correct answer is the one that is different. For example, if there are three plural verbs and one singular verb, the singular verb is the correct answer.

Pro Tip: It/They Test

If you’re not sure whether a verb is singular or plural, try the “it/they test.” Try to use the verb after “it” or “they.” If it makes sense with “it,” then the word is singular. If it makes sense with “they,” then the word is plural. If both make sense, then it’s a neutral word.

Example: “It serves” versus “they serves”: “It serves” sounds better, so “serves” is singular.

If you have doubts, check back to the sentence and identify the subject. Hopefully, you will see that the subject matches the verb perfectly. Let’s try an example:

During the American Civil War, Thomas Morris Chester braved the front lines as a war correspondent for the *Philadelphia Press*. Amplifying the voices and experiences of Black soldiers _____ of particular importance to Chester, who later became an activist and lawyer during the postwar Reconstruction period.

Which choice completes the text so that it conforms to the conventions of Standard English?

- A. were
- B. have been
- C. are
- D. was

Subject/Verb Crusher: Answer choices (A), (B), and (C) are all plural, while (D) is singular. That's the right answer! No need to use our thinking parts.

Check and Verify (Optional): In this case, what is of particular importance to Chester. Careful! It's not "the voices and experiences," or "the Black soldiers." It's "**Amplifying...**" those things. The gerund form of a verb is always singular, so the singular answer makes sense.

Important Note: If any of the answers does not fit the plural/singular pattern, the question is not subject/verb agreement. For example, an answer like "will have" is not singular or plural, because it works for both cases. If you see an answer like this, do not attempt subject/verb agreement. It's probably a question about verb tense.

Reading Writing Skill 7: Understand Semicolon Use

A semicolon (;) is a punctuation mark that is used to indicate a stronger separation between independent clauses than a comma, yet a weaker separation than a period. Here are the primary ways in which a semicolon is used:

Between Independent Clauses:

A semicolon can be used to connect two closely related independent clauses (complete sentences) that are not joined by a coordinating conjunction (such as "and," "but," "or," etc.).

Example: The weather was sunny; people were enjoying outdoor activities.

In a Series with Commas (Supercomma):

When listing items in a series where individual items already contain commas, a semicolon can be used to separate the items. These semicolons, often called “**supercommas**,” help to prevent confusion between the items in the list. The most obvious clue that a sentence contains a supercomma list is the presence of “; and” somewhere in the main clause. Only a supercomma list would use a semicolon with a conjunction.

Example: My favorite cities are Paris, France; Rome, Italy; and Barcelona, Spain.

Example: The conference attendees included Jane Doe, CEO of XYZ Corp.; John Smith, Chief Scientist; and Dr. Robert Johnson, Professor of Physics.

Note: The presence of a conjunctive adverb (a transition word like “however” or “therefore”) does not remove the need for a semicolon between two independent clauses. Using a transition word in this context creates a **comma splice**, a common grammatical error.

Reading Writing Skill 8: Know the Rules for Colon Use

A colon (:) is a punctuation mark with several uses in writing. Most students have learned a limited or misleading definition of the colon that will not work for the SAT. Here are the two rules you need to know:

1. A colon can only be placed after an independent clause.
2. A colon **explains** or **clarifies** whatever came before it.

In general, colons are used to introduce whatever information or justification follows. Here are the most common cases:

Introducing a List or Series:

Use a colon to introduce a list or series of items.

Example: "She bought three items at the store: milk, bread, and eggs."

Introducing an Explanation or Example:

A colon can be used to introduce an explanation or example that elaborates on the preceding statement.

Example: "There was only one way to describe the feeling: pure exhilaration."

Introducing Quotations:

Use a colon to introduce a quotation or a block of text.

Example: "The famous quote by Albert Einstein is true: 'Imagination is more important than knowledge.'"

Between Independent Clauses (with Emphasis):

A colon can be used to connect two independent clauses when the second clause emphasizes or explains the first.

Example: "The solution is simple: work hard and stay focused."

A good rule of thumb is that the independent clause in these cases feels like a “cliffhanger,” meaning you get the sense that some important detail is being withheld on purpose. Then whatever comes after the colon will resolve the cliffhanger and give you a sense of closure. Once you get the hang of colons, there’s only one way to describe these questions: easy!

Reading Writing Skill 9: Know the Rules for Using an Em-Dash (—)

An em-dash (—) is a punctuation mark that is longer than a hyphen and is used to set off or emphasize information in a sentence. When used inside a sentence, the information is placed inside two matching em-dashes. When used at the end of a sentence, only one em-dash is needed, because the period terminates both the insertion and the sentence as a whole.

Here are some common uses of the em-dash:

Emphasizing or Setting Off Information:

Use em-dashes to set off information that is parenthetical or adds emphasis. It's a more dramatic break than parentheses or commas.

Example: "The chef—known for her signature dish—won the competition."

Interrupting Thoughts:

Em-dashes can be used to indicate an abrupt change or interruption in thought.

Example: "I was about to leave—wait, did you hear that noise?"

Adding information to a complete thought:

Em-dashes can be employed to insert information at the end of a sentence, either to contrast or expand the preceding information.

Example: "There was no one more suitable to working as a Walmart greeter than Jimmy—no one, that is, except his sister Ginny."

When using em-dashes, it's important to note that there should be no spaces on either side of the em-dash. In word processing software, you can often create an em-dash by typing two hyphens (--) and letting the software automatically convert them into a single em-dash.

Reading Writing Skill 10: Commas

A comma (,) is a punctuation mark that is commonly used in writing to indicate a brief pause or separation of elements within a sentence. Commas serve various purposes in sentence structure and are essential for clarity and proper interpretation.

What commas can do:

Separating items in a list:

Commas are used to separate items in a list of three or more.

Example: "She bought apples, bananas, and oranges at the grocery store."

Separating independent clauses (with a coordinating conjunction):

Commas are used to separate independent clauses (complete sentences) when they are joined by coordinating conjunctions (and, but, or, nor, for, so, yet).

Example: "I went to the store, and I bought some groceries."

Setting off an introductory clause or phrase:

Commas are used after introductory words, phrases, or clauses at the beginning of a sentence.

Example: "However, I decided to go to the party."

Setting off nonessential information:

Commas are used to set off nonessential or nonrestrictive elements within a sentence.

Example: "My friend, who lives in Paris, is coming to visit."

Separating adjectives:

Commas are used to separate coordinate adjectives (adjectives that equally modify the same noun).

Example: "It was a beautiful, sunny day."

In the SAT Writing, commas are only used when absolutely necessary, meaning each comma in a sentence must have a specific purpose. If you're not sure why you would place a comma in a particular spot, **leave it out**.

What commas CANNOT do:

Connecting two independent clauses:

A single comma is not strong enough to connect two independent clauses. Using a comma in this forbidden scenario is a common grammatical error called a **comma splice**. You can fix a comma splice by adding a **coordinating conjunction** (and, or, but, etc) or replacing the common with a period, a semicolon, a colon, etc.

Wrong: "I'm hungry, I would like a cheeseburger." (comma splice!)

Correct: "I'm hungry, so I would like a cheeseburger."

Separating a noun from its verb, or separating a verb from its object:

The natural order of an English sentence is Subject, Verb, Object (Example: Johnny ate a bunny.) You should not place a single comma between these essential sentence components, even if it feels like a good place for a pause.

Wrong: "The girl waiting at the bus stop for her beloved European foreign exchange student boyfriend, was crying uncontrollably."

Right: "The girl waiting at the bus stop for her beloved European foreign exchange student boyfriend was crying uncontrollably."

Separating a noun from an *essential appositive*:

An **essential appositive** is a noun or noun phrase that renames another noun and is necessary to identify it. Because it's essential to the meaning of the sentence, it is not set off by commas.

Wrong: "Entomologist, Steve Buggerson, was named Man of the Year by Time Magazine."

Right: "Entomologist Steve Buggerson was named Man of the Year by Time Magazine."

Other examples of essential appositives: "The novel Beloved..." "The river Mississippi..." "The chemical compound NaCl..."

Note: For all punctuation questions, but particularly for comma questions, it's very important to **read the entire sentence** before deciding what punctuation is appropriate. It often happens that a particular punctuation mark seems quite correct locally, but creates a conflict with a punctuation mark or structural element somewhere else in the sentence.

Reading Writing Skill 11: Punctuation Mark Strategy

Even if you understand the basic use of each punctuation mark, it can still be challenging to navigate the questions in the Boundaries topic. For one thing, many wrong answer choices are deliberately designed to fool your “Error Alert,” that feeling of wrongness you get when you see an incorrect sentence. Other wrong answer choices are grammatically correct near the underlined part, but contain an error farther down the sentence.

You can avoid many traps by checking the answer choices in this very specific pattern:

- 1) **“;” “.” “, conjunction”** Always start with the answer choices that offer a semicolon, period, or comma+conjunction.
 - a) Does the list require a “supercomma”? Look for the presence of “; and” somewhere in the sentence, which indicates that the items in the list are being separated by semicolons. If not.... Proceed to the next step.
 - b) If two answer choices are identical, except one uses a semicolon and one uses a period, **eliminate both answers**. If not, move on the next step.
 - c) For the semicolon, period, or comma+conjunction, is there an independent clause before the punctuation mark? Is there an independent clause after the punctuation mark?
 - d) If the answer to these questions is “Yes,” then you need a semicolon, period, or comma+conjunction.
- 2) **“:”** Next, check the answer with a colon.
 - a) Is there an independent clause before the colon?
 - b) Does the independent clause feel like a “cliffhanger”?
 - c) Does the clause after the colon (dependent or independent, doesn’t matter) help **explain** or **clarify** the independent clause?
 - d) If “Yes,” then the colon is correct.
- 3) **“—”** Next, check the dashes.
 - a) If an answer creates a pair of two dashes, are the dashes surrounding a nonessential detail? Can the sentence stand alone, without the information, and still say basically the same thing? If “Yes,” choose the pair of dashes.
 - b) If an answer contains a single dash, is the dash located after an independent clause? Is the dash followed by a nonessential detail that ends the sentence? If “Yes,” choose the single dash.

If all else fails.... Proceed to comma placement.

Reading Writing Skill 12 Utilize Transition Phrases

Transition phrases are words or phrases that connect ideas, sentences, or paragraphs to enhance the flow of writing. They help guide the reader through the text, indicating relationships between different parts and ensuring a smooth progression of thoughts.

To choose the correct transition phrase in an SAT question, follow this process:

1. Read the sentence located immediately before the sentence containing the transition phrase.
2. Determine the relationship between two sentences (do not look at the answers yet).
3. Look for an answer choice that matches the relationship you determined.
4. If there are no answers that match, try each one in the sentence. Eliminate answer choices that are too similar to each other to be distinguishable (however vs nevertheless, etc).

For this process to work, it's very important that you consider the precise definition of each transition word, not the gist or general idea. The transition words in the SAT are as precise as mathematical formulas. Here are the most common ones and their SAT-required meanings:

Transition Word	Meaning
however, conversely, on the other hand, by contrast, besides, on the contrary	Expresses a contrast
for instance, for example, specifically, in particular	The second statement is an example of the first statement
in addition, similarly, furthermore, in comparison, additionally, likewise, moreover	The second statement is another item or statement similar logically to the previous one.
indeed, in fact	The second statement confirms and intensifies the first statement
therefore, thus, hence, as a result, consequently, for this reason, fittingly, to that end	The first statement causes or explains the second statement
ultimately, finally, secondly, firstly, lastly, currently, today, next, subsequently	Indicates the respective step or time in a sequence
granted, of course, admittedly, that said	The second statement provides a qualifier that reduces the significance of the first statement

nevertheless, regardless, still	The second statement is true despite the first statement
that is, in other words	The second statement restates the first statement, in simpler language
instead, alternately, alternatively	The second statement provides a different take or approach from the first statement

Note 1: A transition phrase can be located anywhere in a sentence, even the very end; no matter where it's placed, a transition phrase always connects to the **previous sentence** or **independent clause**.

Note 2: Some transition phrases don't necessarily connect the two sentences, but instead move the story along to the next step or point in time.

Example: "I went to the store to search for a square watermelon. Eventually, I was forced to give up my quest. "

Reading Writing Skill 13: Possessive Cases

In grammar, the **possessive case** is a construction that indicates ownership or possession of something. When a noun or pronoun appears in the possessive form, it possesses whatever noun or noun phrase immediately follows.

Here are the basic rules for forming the possessive case:

Possessive Nouns

Noun possessive forms always use an apostrophe ('), but vary in the placement of the apostrophe and in whether they need an "s".

1. Singular Nouns:

For singular nouns, add an apostrophe followed by "s."

Example: The dog's tail (indicating that the tail belongs to one specific dog).

2. Plural Nouns Ending in "s":

For plural nouns that end in "s," add only an apostrophe.

Example: The students' notebooks (indicating that the notebooks belong to multiple students).

3. Plural Nouns Not Ending in "s":

For plural nouns that do not end in "s," add an apostrophe followed by "s."

Example: The children's toys (indicating that the toys belong to multiple children).

Possessive Pronouns

Some pronouns also have possessive forms, but they **do not** require an apostrophe. For example, pronouns such as "his," "her," "your," "its," "their," "whose," etc. do not use apostrophes, but still indicate possession.

Joint Possession

When multiple people share ownership, only the last noun receives the possessive marker.

Example: John and Mary's car (indicating that the car belongs to both John and Mary).

Reading Writing Skill 14: Match Pronoun and Antecedent

A pronoun is a word that refers to a noun that comes before it (or, in some cases, after it). The antecedent is the word or group of words to which a pronoun refers.

For example:

- Mary bought a new dress. She wore it to the party.
- The noun "Mary" is the antecedent.
- The pronoun "She" refers back to "Mary."
- The pronoun "it" refers back to the dress.

Here are some basic rules to keep in mind for SAT and ACT pronoun questions:

1. It's important for the antecedent and the pronoun to agree in gender and number. In English, pronouns such as "he," "she," and "it" refer to singular nouns, while "we" and "they" refer to plural nouns.
2. When the gender of a singular noun is not known, use "he or she" or "him or her." On the SAT, the pronoun "they" is always considered plural.
3. Every pronoun should have a clear, unambiguous antecedent. There should not be another "candidate" for the antecedent that is approximately the same distance (or closer) to the pronoun. For example: "John and Mark both ate the lobster, but it was a bad choice because he was allergic." Who has the shellfish allergy?

Reading Writing Skill 15: Dangling Modifiers

In grammar, a modifier is a word, phrase, or clause that provides more information about another element in a sentence. Modifiers play a crucial role in enhancing the meaning of nouns, verbs, adjectives, or other adverbs. They help to clarify, describe, or limit the meaning of the word they modify. **Modifiers must be placed next to the phrases they modify.**

There are several types of modifiers:

1. **Adjectives:** These modifiers describe or limit nouns and pronouns. They answer questions such as "what kind?" or "which one?"
Example: The tall man waved.
2. **Adverbs:** Adverbs modify verbs, adjectives, or other adverbs. They often answer questions such as "how?" "when?" "where?" or "to what extent?"
Example: She spoke softly.
3. **Phrases and Clauses:** Modifying phrases and clauses can act as adjectives or adverbs. They add information to nouns, pronouns, verbs, adjectives, or adverbs.
Example: The house with the blue door is mine. (Adjective phrase modifying the noun "house")

Dangling Modifier

A dangling modifier is a word or phrase in the introductory clause position that describes a noun phrase, but the noun phrase it describes is missing from the sentence or not placed in a logical position.

Here are a couple of examples:

Incorrect: Running late, the train was missed. (Who is running late?)

Correct: Running late, he missed the train.

Incorrect: Having finished my homework, the TV was turned on. (Who turned on the TV?)

Correct: Having finished my homework, I turned on the TV.

To avoid dangling modifiers, you must place the word or phrase being modified immediately next to the modifier.

Reading Writing Skill 16: Subject vs Object Pronouns

In English, most pronouns have separate subjective and objective forms. In general, we use these forms correctly without conscious effort, but there are borderline cases where the correct pronoun form is less obvious. First, some basic terms:

- **Subject Pronouns** are used as the subject of a sentence or clause and perform the action of the verb. Examples: I, he, she, etc.
- **Object Pronouns** are used as the object of a verb, **receiving** the action. Examples: me, him, her, etc.

Here are the most common English pronouns in both forms:

Subject	Object
I	me
he	him
she	her
they	them
we	us
you	you
who	whom

To decide between the two pronoun forms in one of the borderline cases, a useful tactic is to do a **replacement**, changing the nouns and pronouns of the sentence to a simpler case. Check whether the new sentence uses object or subject pronouns, then apply the same case to the original sentence:

- My parents bought (**my sister and me/I**) a smoothie. My parents bought **me** a smoothie. My parents bought my sister and **me** a smoothie.
- Between (**you and me/I**), chocolate is evil. Between (**us**), chocolate is evil. Between you and **me**, chocolate is evil.
- (**Who/whom**) just called the house? **He** just called the house. **Who** just called the house?

Note that in the third example, we changed the question into a declarative statement for extra simplicity. As long as you keep the general relationships in the sentence the same, you can make adjustments to tense, mood, plurality, etc.

Reading Writing Skill 17: Research Notes Questions

At the end of each Reading/Writing module, you'll find a group of questions about research notes. They usually begin with, "While researching a topic, a student has taken the following notes...." These questions are quite straightforward, but it's important to focus your attention correctly:

1. **Start by reading the question, but note that the important information is actually the first part of the question, not the second.**

There is always a statement that says something like, "The student wants to...." This is the heart of the question. Your job is to find the answer that does whatever the student intends to do, as specifically as possible to the student's goals. The next statement is always something like, "Which choice most effectively uses...." That statement is the same in every question of this type, so it's not useful information.

2. **Do not read the notes themselves.**

The answer choices are nearly always correct summaries of the notes, so it's not your job to verify the facts.

3. **Find the answer that specifically accomplishes the student's goal.**

If you're stuck between two answer choices, check the student's goal again. Is one of the answer choices slightly off focus, or missing a key element? Note that the SAT uses the word "and" very precisely. If the student's goals include two phrases connected with "and," it means there are two requirements for the correct answer, not one. Is one of the answer choices doing only half of what the student wants? There's almost always something wrong with three out of the four answer choices.

4. **If two answers both seem to be supporting the student's goal, check the facts.**

In the very rare cases where multiple answers fulfill the requirements of the question, you then have to check the facts in those answers and verify that they match the research notes. Roughly 5% of research questions have wrong answers that are misinterpretations of the data.

Here are some common phrases in the “student wants” sentence that connect to specific requirements:

Phrase	The correct answer will...
“Introduce a study”	Mention the scientist, the general goal of the study, the year it was conducted, etc.
“Present the results of a study”	Specifically provide the data produced by the study or experiment
“Present the conclusions of a study”	Include the interpretation or analysis of the study results, essentially what the results “mean”
“Generalization about X”	Contain a broad statement that describes only X, but not necessarily anything else in the notes.
“Specific evidence for X”	Contain a factual example that supports X
“Indicate X”	Include the specific fact mandated by X
“Contrast” or “compare” “Emphasize a different” or “emphasize a similarity”	Information about two different things, to indicate how they are either different or similar
“An audience unfamiliar with X”	A summary or brief explanation of X,
“An audience already familiar with X”	Nothing that identifies or explains X; instead the answer will use X as though everyone knows exactly what it means.
“Emphasize how X does Y”	Include the specific method by which X accomplishes Y. It’s not enough to only say “X does Y”

Reading Writing Skill 18: Inference Questions

A small number of SAT Reading questions ask students to make **inferences** (logical conclusions) from facts or data presented in a passage. The key to these questions is to remember that, even when you're explicitly asked to infer or draw conclusions, you should still focus primarily on the information already stated. In other words, don't infer too much!

You can recognize inference questions by the phrase they all contain: "Which choice most logically completes the text?"

Example:

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

- A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.
- B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.
- C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.
- D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

Strategy:

Like most reading questions, some version of "Good Cop, Bad Cop" will bring you closer to the correct answer:

1. Read the passage carefully, summarizing each sentence (or each part of a sentence, when things get complicated) into a brief "soundbite." Make sure your soundbites are capturing the essence of the sentences, rather than focusing on particular details.
2. Next, ask yourself, "If I know those things are true, what else is also obviously true?" Try not to focus on the facts separately, instead thinking about what happens when you put the facts *together*. Make sure you take a little information from each sentence.

3. Next, consider each answer choice and ask, “Is this answer completely supported by the passage? If the above statements are true, do I absolutely *know* this is true?” Eliminate any answers that are not 100% supported by information that is already stated.
4. For each answer choice, ask, “Does this answer put information together from two or more places in the passage?” Eliminate any answers that do not refer to multiple statements in the passage.

If you’re careful with these steps, you should only be left with one correct answer—the one that connects facts in the passage together, without adding any new details or assumptions.

The wrong answers will bring up information or connections that are not provided in the passage, requiring one or more assumptions. Sometimes, the wrong answers are very reasonable statements, probably true in most cases, but not directly supported. That still means they’re wrong!

Essential Skills for Math

The SAT Math section typically covers concepts that almost all students have learned by the time they complete Algebra 2. However, SAT math questions ask students to apply these concepts in ways that are unfamiliar or confusing. They tend to emphasize the borderline cases or force students to engage with the “why” underneath familiar math topics.

Annoying, right? Most students treat math as a set of recipes or formulas: when I see *this* kind of problem, I do *these* steps. To be successful on the SAT, you have to get more flexible in your thinking.

This guide covers a series of skills that are useful on the SAT. Don’t memorize the steps. Instead, consider why each step makes sense. Why are you even allowed to do that step? Learn to recognize when certain strategies work.... And why they work.

As you do more and more SAT practice questions, you’ll start to see patterns that will help you interpret questions in the future. However, you’ll never see 100% of the possible SAT questions, because each test always brings new questions and challenges. The only way to prepare for that is to be comfortable with discomfort. Follow these basic principles:

- 1) **Read** every question carefully. Don’t just skip to the numbers or the equations. There is often valuable or even crucial information in the text of math questions.
- 2) Focus on **definitions**. Each math term in the SAT has a specific definition, and you can use that definition to help you decide how to incorporate that term into your process.
- 3) Decide consciously whether or not to use **Desmos**. Don’t default to always using the calculator or always using algebra. There are questions that favor each of those approaches, so make the decision deliberately at the beginning of the question.
- 4) Learn to identify **question types**. As we said earlier, there are patterns in the SAT Math that you can use to your advantage. Learn to look for the keywords or other elements that signal certain question types. This guide contains many of them.
- 5) **Try stuff**. When you feel frozen at the start of a question, take a step forward by doing... stuff. Combine like terms. Simplify. Plug in your own numbers. Taking a few steps can help the question make more sense in your mind, and it might even lead all the way to an answer.

Now, we’ll present you with some of the most useful strategies you can use on the SAT Math. Remember: don’t memorize steps! These skills are **tools** in your **toolbox**. Don’t try to use a wrench where a hammer would work better.

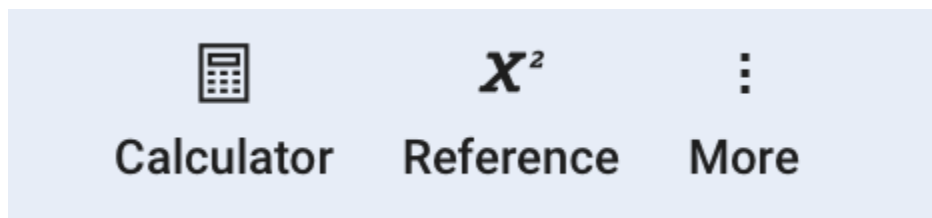
Math Skill 1: Desmos, Your New Best Friend

The Digital SAT (DSAT) allows students to use a calculator for all questions in the Math section. Almost any calculator is permitted, including the scientific or graphic calculator that you're probably using in your school math class.

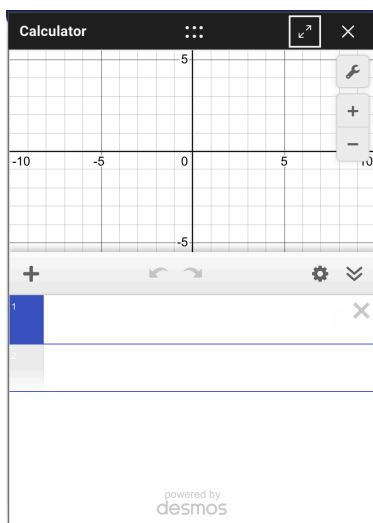
However, the test also provides a built-in calculator powered by Desmos, and that calculator is much faster and more useful than any calculator you might bring with you. In this guide, I'll walk you through the most important tools available in the Desmos calculator, as well as the most common question types where Desmos can make a difference.

Lesson 1: Where's My Desmos?

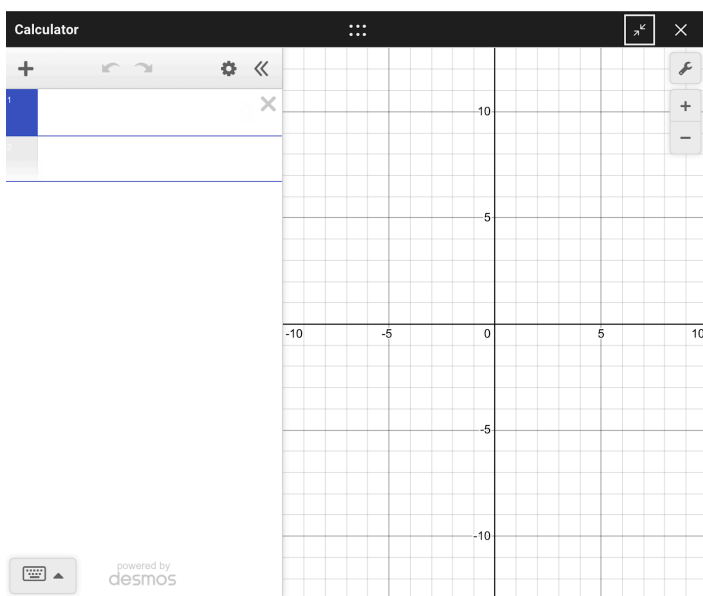
So where is this magical calculator? When you start an SAT Math section, the calculator and reference sheet may be hidden by default. At the top right of your first math question, you will see these buttons:



Click on the Calculator button and Desmos will appear on your screen. Voila! Now, when it first opens, Desmos will probably be in its compact mode, which looks something like this:



This version of Desmos works well if your testing device has a smaller screen, but I would recommend you hit the “Expand” button at the top right of the calculator. You’ll get the much roomier and more useful full-size mode of the calculator, which looks like this:



Lesson 2: Basic Calculations

To do some of the typical operations that human beings invented calculators to do (multiplication, square roots, etc), you can simply type your expression into the selected line next to the graph

screen. The lines are numbered, so you will likely type your first expression into Line 1, shown highlighted with blue in the picture above.

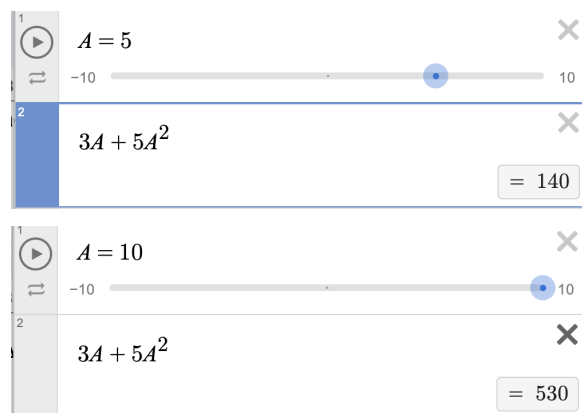
Desmos understands many keyboard characters and basic shortcuts, and it's worth your time to memorize most of these:

- Multiplication: *
- Fraction: /
 - Type / before your first number to create a fraction with a numerator and denominator
 - Type / after you've already entered a number to turn your input into a fraction, with your inputted number as the numerator
- Absolute Value: |
 - Located over the backslash button, just above the Enter key
- Exponent: ^
 - Located over the number 6 on a standard keyboard
- Square root: sqrt()
 - Desmos will convert the shortcut into an actual square root symbol
- Percent: %
 - Located over the number 5. Add the word "of" and Desmos will find percents for you. Example: 20% of 30 —> Desmos output will be 6

Lesson 3: Assigning Names and Other Shortcuts

Have you ever had to plug a number into a function, tediously writing out every step on your calculator screen, then discovered that you actually need to do the whole thing again with other inputs? Desmos knows that your life is already difficult enough, so it offers many convenient features that allow you to save information for later use.

- Assign a value to a variable
 - Type $A = 5$ (Or whatever variable / number you need). From now on, every time you type the letter A in another line, Desmos will use the value 5 for A.
 - Use assigned values to try many numbers in an expression. Assign a value to A, then write the expression out on another line, then change the value of A as much as you want. Like this:



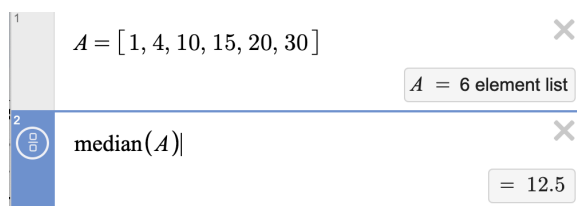
- Use function notation to name and utilize functions
 - Type $f(x) = 3x^2 + 5$. From now on, when you type $f()$ with an input in the parentheses, such as $f(3)$, Desmos will use the definition of $f(x)$ to evaluate the function.
 - If you type $f()$ with an expression in x for an input, Desmos will substitute the expression in the original equation and then graph the result.
 - You can even define new functions using the definitions of pre-existing functions. For example, you can write $g(x) = f(x) + 3$, and Desmos will create a function named $g(x)$ that is identical to $f(x)$, but translated up 3 units on the xy -plane.
- Use variables to store points and lists (you'll see why....)
 - If you need to store a coordinate point or a whole list of numbers (or even a list of points...) for later use, Desmos is perfectly happy to let you do that.
 - To store a point, type $P = (3, 4)$ (or whatever name and point you wish)
 - To store a list, type $P = [1, 3, 4, 10, 40]$ (or whatever name and list you wish). Note that the items have to be separated by commas.

Lesson 4: Advanced Calculator Functions

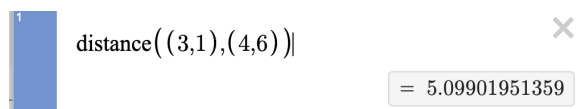
If you click on the keyboard icon, then click Functions, you'll see that Desmos offers a long list of math functions for your calculating enjoyment. I'll run down the most useful ones, but I encourage students to explore the menu on their own and see what other tools might be hiding there.

- Statistics: mean, median, min/etc
 - To use these data set functions, type the name of the function, then a left parenthesis (, then a list of numbers separated by commas, and finally a right parenthesis).

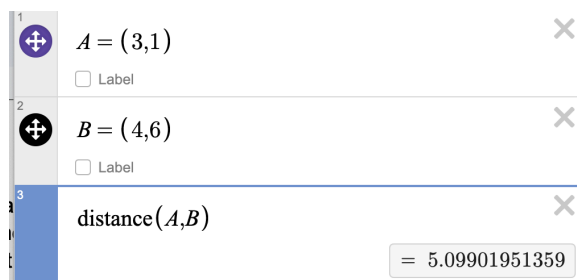
- Better yet, store the list in a variable name, then type the function with the list name inside. Like this:



- Geometry: distance, midpoint, etc
 - To use these coordinate functions, type the name of the function, then a left parenthesis (, then two points separated by a comma (make sure to use parentheses around each point), then a right parenthesis). Like this:



- Better yet, store each point in a variable name, then type the function with the two points inside. Like this:



Lesson 5: Solving Questions with Desmos

The Desmos calculator is a bit like a very nice, expensive hammer. It's incredibly good at specific tasks, but you shouldn't just whack at things with the hammer and hope to do something useful. Before you bring Desmos into a problem, make sure you know (or at least suspect) that the calculator can actually be helpful in that problem.

Here some ways that Desmos can be used on the DSAT:

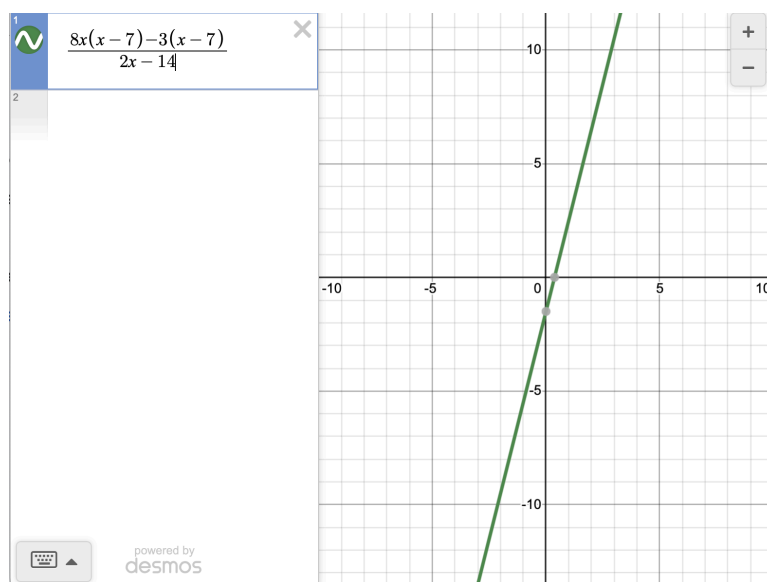
- **Finding Visual Information**

- To identify or check features of equations that have a **visual component**, use the basic graph features.
- Some DSAT questions ask you to find features of an equation that might be easy to identify on its graph, such as the minimum, maximum, x-intercept, y-intercept, etc. Instead of finding these values with algebra, simply graph the equation and find the appropriate value.
- You can also use the graphing utility to check whether two equations are **equivalent** (meaning they are identical). Graph both equations; if they are equivalent, the two graphs will be right on top of each other, to the point where one of them hides the other. So, if a DSAT question asks you to identify an expression that is equivalent to the original expression they give you, simply graph the original expression and then graph each answer one by one. Here's one:

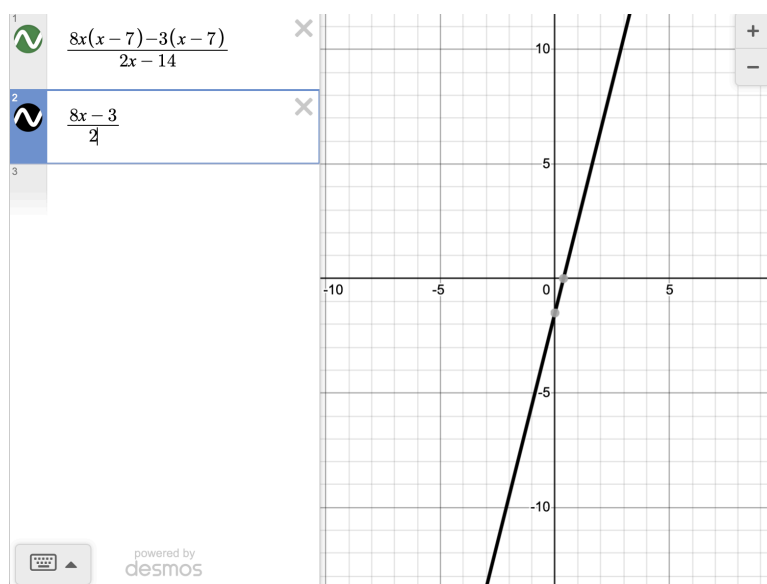
Which expression is equivalent to $\frac{8x(x-7)-3(x-7)}{2x-14}$, where $x > 7$?

- A. $\frac{x-7}{5}$
- B. $\frac{8x-3}{2}$
- C. $\frac{8x^2-3x-14}{2x-14}$
- D. $\frac{8x^2-3x-77}{2x-14}$

Yikes! But don't despair: simply graph the original equation, then graph each of the answer choices one by one, like this:



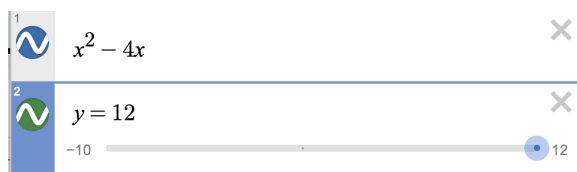
When you graph the right answer, you'll see both graphs will overlap each other completely:



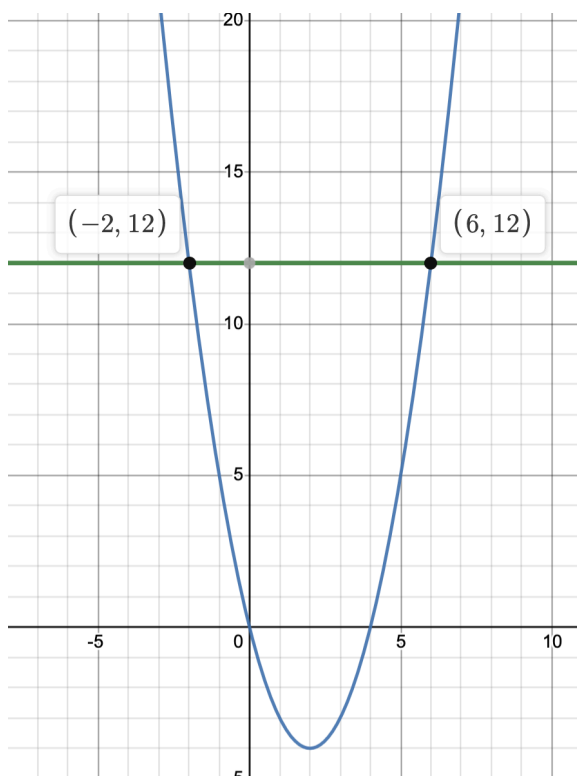
Where did my green line go? Oh, right, it's been perfectly covered by the new graph, so B is the right answer.

- **Solving Equations**

- To solve an equation: when a question requires you to solve a basic equation in any variable, whether as the main task or as one step along the way, you can use Desmos to immediately find the solutions.
- **Method 1:** Type each side of the equation into a separate line in Desmos. You should see the graphs of the two expressions appear on your graph screen.
 - If your equation uses a variable other than x, replace the variable with x. If one side of the equation is simply a number, add y= before the number to turn the number into a graphable equation.
 - Look for the point of intersection of the two graphs. If you click on either of the graphs, Desmos will create a gray point at any intersection point. Huzzah!
 - For example, to solve $x^2 - 4x = 12$, type each side into a separate line like this:

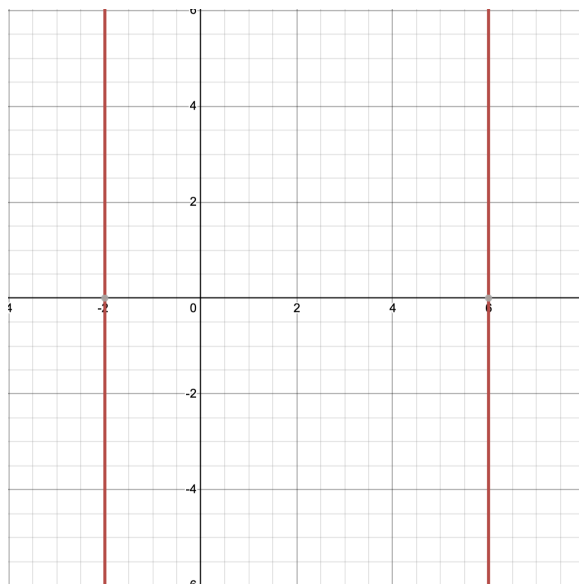


Look for the intersection points in the graph:



- **Method 2:** Simply type the entire equation into a single line in Desmos, including the equal sign. Desmos will graph the solutions as vertical lines.
 - When you type an equation in x into Desmos, you will not see the equation of the graph. Instead, each solution ($x = __$) will create a single vertical line on the graph. This is because Desmos will solve the equation numerically, then treat each solution as a separate graph; since any equation in the form “ $x = __$ ” is a vertical line, the solutions will appear as vertical lines.
 - Example: to solve $x^2 - 4x = 12$ using this method, simply type the equation into Desmos, like this:

- You should immediately see two vertical lines appear on your graph. If you don't see them, you might need to adjust your zoom or window to align your graph.



Each vertical line represents a solution. Here, we can see $x = -2$ and $x = 6$ are solutions to the equation.

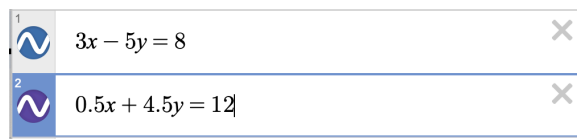
● Systems of Equations

- To solve a system of equations, use the same basic process as above, with some small modifications.
- Type each equation entirely into a separate line. No need to split the equations.
- If the variables in the equations are not x and y , convert them to x and y . Make a note of which variable you replaced with x and which with y , so you don't mix up the answers later.
- Do not simplify or make any changes whatsoever to either equation. Desmos can graph an equation in any form, so it's better to keep your own steps to a minimum to avoid wasting time or making errors.
- Look for the point of intersection between the two graphs. Note that there may be more than one intersection point, in which case each point represents a different solution.
- For example, to solve a system like this:

$$3a - 5b = 8$$

$$0.5a + 4.5b = 12$$

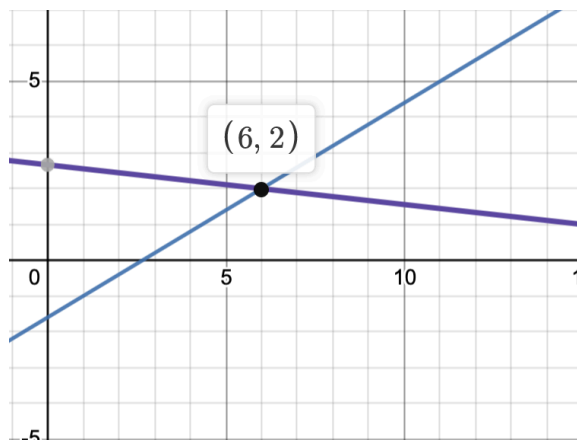
Convert the variables to x and y, then plug into Desmos:



1 $3x - 5y = 8$

2 $0.5x + 4.5y = 12$

Now look for the point(s) of intersection:



Success! $a=6$ and $b=2$

• Sliders

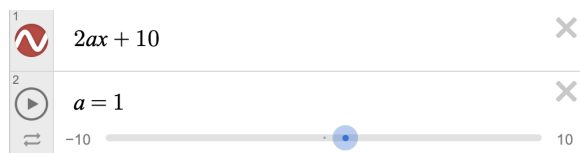
- To try many possible values for constants in an equation, use sliders.
- Some questions on the SAT involve the graphs of equations that have one or more **constants** represented by letters.
- Type the equation into Desmos, including the constant. As soon as you type the constant, Desmos will ask if you wish to add a slider:



1 $2ax + 10$

add slider: a

- Click on the blue box around the letter, and you will see a slider appear:



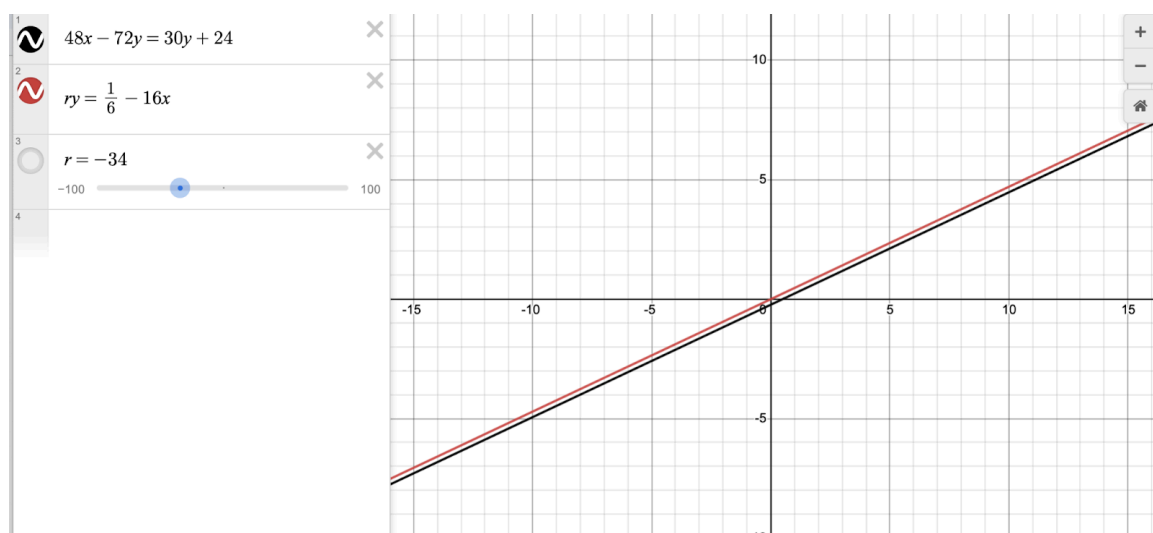
- If you hold down the slider and move it left to right, you will see the graph changes shape with each new value. You can also click on the value of the constant and manually enter any number you wish, as well as set the minimum, maximum, and step (how much the value changes as you move the slider around) for the constant.
- Sliders are great tools for questions that ask for the value of a constant that creates a specific scenario:

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

Graph both equations in Desmos, adding a slider for the constant r . Then, move the slider around until you can see that there is no solution (the lines will not intersect):



● Regression from Points - Basic Regressions

- To find the standard form of an equation, given two or more points in the equation, use the **regression** tool.

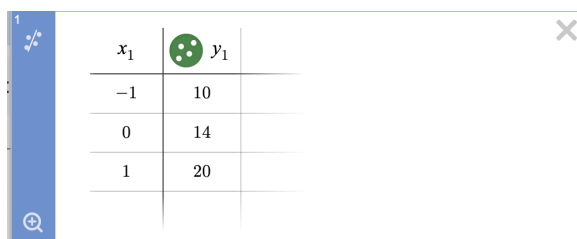
- When you have two or more points that lie on a standard equation type (linear, quadratic, cubic, etc), you can use one of the built-in regression types. Let's take a look at an example:

ID: 44076c7d

x	$f(x)$
-1	10
0	14
1	20

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = 3x^2 + 3x + 14$
 - B. $f(x) = 5x^2 + x + 14$
 - C. $f(x) = 9x^2 - x + 14$
 - D. $f(x) = x^2 + 5x + 14$
- First, make a table with the points. Click on the little plus sign on the top left of the Desmos input column (the tool tip says "Add Item" when you hover over the button), then select Table. Add the three points from the question into the x_1 , y_1 columns:



x_1	y_1
-1	10
0	14
1	20

- Next, look for the regression symbol on the left side of the tile: . Press it....
- Immediately, you should see your tile expand to include a drop-down menu that reads "Linear Regression." Below that menu, an equation should have appeared. That equation is the line of best fit for the data in your table.
- However, we don't want an equation of a line; the question asks for a quadratic equation, and that's what we need to find. Click on the drop-down menu and select "Quadratic Regression."
- Presto Changeo, the equation should now correctly be quadratic:

Quadratic Regression

EQUATION

$$y = 1x^2 + 5x + 14$$

STATISTICS RESIDUALS

$R^2 = 1$ e_1

- That equation is the answer to the question, matching option (A).

● Regression from Points - Custom Regressions

- To find the constants in an equation that doesn't fit one of the standard regression types, given two or more points in the equation, use the **regression symbol** ($\sim$).
- Let's say a DSAT question gives you two or more points for an equation, then tells you something like “the equation can be written in the form....”, followed by an equation with a number of constants. Statistical regression is used to find an equation that best matches a set of data points, but when the points are actually on the graph, regression leads to the exact equation that goes through those points.
- Here's an example of a question that fits this template:

The function f is defined by $f(x) = a^x + b$, where a and b are constants. In the xy -plane, the graph of $y = f(x)$ has an x -intercept at $(2, 0)$ and a y -intercept at $(0, -323)$. What is the value of b ?

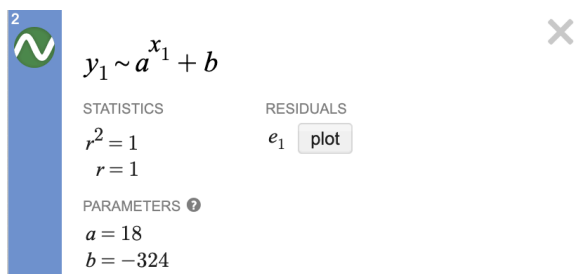
Notice we have two points— $(2, 0)$ and $(0, -323)$ —and an equation with some constants in it. A perfect candidate for regression.

- First, make a table with the two points. Click on the little plus sign on the top left of the Desmos input column (the tool tip says “Add Item” when you hover over the button), then select Table. Add the two points from the question into the x_1, y_1 columns:

x_1		y_1
2		0
0		-323

- Now, in the next line, type the equation you're trying to find, **replacing x and y with x_1 and y_1** , and **replacing the $=$ sign with the regression sign, $\sim$** . Note: to

get the correct x_1 with the subscript, just type x1 into Desmos, which will automatically convert it to the correct format. Your output should look like this:



You will immediately see that a and b now have values ($a=18$, $b=-324$). So the answer to this question is -324. You can ignore all the other stuff.

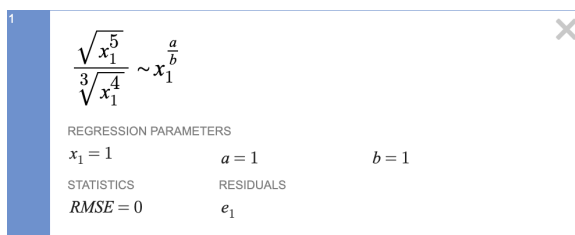
● Regression for a Single Equation Identity

- When a single equation is “true for all x” or something along those lines, it means the left and right side of the equation are **identical**, or equivalent. This kind of equation is called an **identity**.
- If an SAT question contains an identity, but there are constants missing from the equation, you can use regression to find their values.
- Here’s an example:

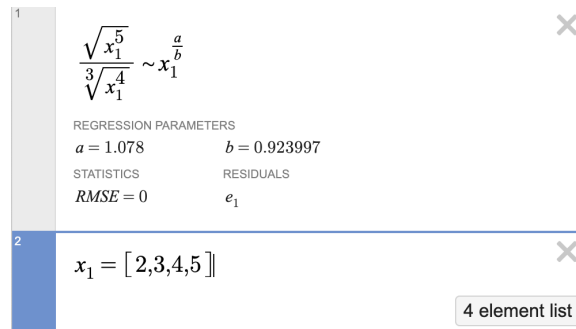
$$\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$$

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x ,
what is the value of $\frac{a}{b}$?

- Start by typing the equation into Desmos exactly as you see it, but making the same replacements we explained earlier: “x” → “x1”, “y” → “y1”, and “=” → “~”.



- Notice that Desmos has already decided what the values are, including the value for “x,” which it set to equal 1. If you plug these numbers into the equation, you will see that they do, indeed, make the two sides equal. However, we want the values of “a” and “b” that make the equation true for **all x**, not just x=1.
- So, next, set x equal to a few different values (doesn’t really matter which values). Use the square brackets [] to create a list of values. Try to avoid numbers that would cause a division by zero in the original equation (so don’t use x=0 in this case):



1

$$\frac{\sqrt{x_1^5}}{\sqrt[3]{x_1^4}} \sim x_1^{\frac{a}{b}}$$

REGRESSION PARAMETERS

$a = 1.078$ $b = 0.923997$

STATISTICS RESIDUALS

$RMSE = 0$ e_1

2

$x_1 = [2,3,4,5]$

4 element list

- Now you can see that the values of “a” and “b” have changed in the regression equation, becoming some odd-looking decimals. Since the original question was asking for “a/b”, let’s go ahead and type that into the next line:



3

$\frac{a}{b}$

= 1.16666666667

- Then hit the fraction symbol ($\square/\square$):



3

$\frac{a}{b}$

= $\frac{7}{6}$

Conclusion

Think of Desmos as another tool in your toolbox. It's not all-powerful, and it will not magically bring your score to an 800. However, Desmos is quite useful, and anyone who learns to use it well will have an advantage over anyone who's still relying on a physical calculator. It's important to be very familiar with the calculator interface, so you should always use Desmos when you're doing DSAT practice questions or even math homework for school. Become the Desmos master you were always meant to be.

Math Skill 2: Know When and How to Use “My Numbers”

Many SAT problems contain variables that do not have specific values. This typically happens when there isn't enough information to solve for exact values, or when there is a range of possible values for a variable, any of which is equally valid.

In situations where a variable does not have an exact value, one powerful strategy you can employ is to **use your own numbers**. We call this strategy **My Numbers**.

The first step in solving a My Numbers question is to identify it. Here are some signs or features that suggest you can use your own numbers:

1. **There are more variables than equations (remember that equations must have equal signs).**

As a general rule, you need the same number of equations as you have variables. If you have too many variables, it may be possible to solve for some of them, or for a combination of two or more, but you'll never be able to get exact values for all your variables. This means you can use your own numbers to simplify these questions.

2. **There are variables in the answer choices.**

If the answer choices have variables, then in all likelihood you can choose anything you want for those variables. The danger in these problems is that choosing certain numbers (1 or 2 for example) can lead to coincidences in which more than one answer choice works. So make sure you check all the answers! If more than one answer choice works for your numbers, then choose different numbers until you can narrow your choices down to one.

3. **All the information you're given is relative.**

This condition works particularly well in geometry problems where all the clues tell you how measurements are related to each other, but don't offer exact values for any of those measurements.

4. **The question involves a word problem with one or more unknowns, which may or may not have variables assigned to them.**

In these cases, using your own numbers can be a great way to explore the situation in the word problem.

In all My Numbers problems, the general procedure for finding the solution is the same:

1. Substitute your own numbers. Keep in mind that sometimes substituting for one variable automatically dictates values for all the other variables. Make sure you follow all the rules you're given.
2. Identify the question and find the answer, using your numbers.
3. Check each answer choice until you find one that matches your answer. If there are variables in the answer choices, use the same values in those variables that you used for the problem.

Math Skill 3: Know When and How to Use “Their Numbers”

On some math questions, you are asked to solve for a single variable or quantity, and each answer choice is a possible value for that variable or quantity. Instead of trying to solve the question in a straightforward way, one option you have is to try out each answer choice in the original question.

Here are some general signs that you can use the answer choices:

1. There is one super important unknown in the question.
2. The question asks for only one value (not the sum of two values, for example)
3. The question uses the word **“equivalent.”**
4. You are matching data to a graph, data to an equation, or an equation to a graph. Pick a point that you know works and try it in each answer choice.

Note 1: If a question asks for the “largest value that...” or “the smallest value that...”, always start with the largest or smallest value in the answers, respectively. If you start with the largest answer, and it works, then that must be the largest one that works!

Note 2: My Numbers and Their Numbers are only tools in your toolkit. Every math question on the SAT has an “official” solution path that will avoid simply plugging in values. That path may be easier and less time-consuming than using real numbers. However, if you keep these tools handy and available to your brain, you can apply them in cases where the solution path is not clear at all to you, or when the solution you’re imagining seems very complex or tedious.

Math Skill 4: Percent and Percent Change

On the SAT, there are many questions that involve change by a given or sometimes unknown percent. For these questions, it is very important to learn a particular approach to calculating percent change. We call it the **one-step solution**.

Before we review that process, however, here are the basic principles that apply to all percent questions. First of all, students should avoid using proportions or other forms of “is over of.” Instead, use these basic definitions to translate a percent problem into arithmetic:

1. “percent (%)” = divide by 100
2. “of” = multiply
3. “is” = equals
4. “what” = variable or unknown

Example: What is 30% of 80? $\rightarrow x = 0.30 \times 80 = 24$

Percent Increase/Decrease

In many percent problems, a given (or possibly an unknown) value is being increased or decreased by a certain percent. Our task in these questions is not just to find a percent, but to increase or reduce a number by the amount represented by that percent. Follow these steps:

1. Determine the **final percent** after the percent decrease, by adding or subtracting the percent change from 100%, depending on whether we are increasing or decreasing the number. For example, when you reduce a number by 10%, you are left with 90%, since 100% represents the original amount. If you are increasing a number by 10%, you will end up with 110%.
2. Find the **final percent** of the original amount, using the steps covered in the previous section.
3. Great success!

Examples:

1. Let’s say we want to buy an \$80 shirt at Banana Republic, but it’s Black Friday, so the shirt is discounted by 30%. That means we have to take away 30% from \$80, so we’re going to be left with

70% of what we started with. That means that, instead of finding the discount and then subtracting it, we can just find 70% of \$80, and that will be the final price of the shirt:

$$P = 0.70 \times \$80 = \$56$$

2. Now, let's take that same discounted shirt, for the low, low price of \$56. Unfortunately, we have to pay sales tax on that shirt, say 6%. So how do we find the real final price of the shirt? Well, if we add 6% to a starting amount, then we're going to be left with **106% of what we started with.** In decimal form, $106\% = 1.06$. So, to find the final price, we multiply our starting value by 1.06:

$$\text{Final Price} = 1.06 \times \$56 = \$59.36$$

We would recommend that you always do percent change problems using this **one-step solution.** Figure out what the final percent will be after the increase or decrease, then multiply your starting value by that percent. While some SAT percent problems can be done in a variety of ways, a subset of them can **only** be done with the one-step solution.

Note 1: It is very important to make sure you only multiply percents with their associated values. For example, in the case of the 30% discount, that 30% is discounted from the **starting price.** That means that you can't take the final value (\$56) and increase it by 30% to get back to the original value (\$80). Because \$56 is a smaller amount than \$80, taking 30% of that number will also give you a smaller change. In other words, you can't work backwards by using the percent change on the final value. If you don't know the original value, assign it a variable (x, for example) and then set up your calculation like an equation.

Note 2: Some percent change questions don't clearly indicate which number is the "starting value," For example, if the question says something like "Number A is 20% bigger than Number B." In these cases, the number that is the basis of comparison is considered the starting value. That means, in our example, Number B is the starting value, then Number A is the result from increasing B by 20%.

Using Desmos for Percent Problems

The above tools and relationships are the baseline information you need to solve basically all percentage questions, including questions that are deliberately designed to prevent Desmos methods. However, for most (a high percentage of.....) problems in this topic, you can actually use Desmos very effectively.

Finding a Percentage (Desmos)

To do basic operations with percentages on Desmos, you use the % symbol and type the expression as literally as possible.

Example: To find 45% of 600, type that exact expression into Desmos (Note that the word “of” will appear automatically after you input the % symbol):

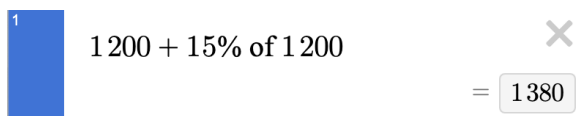


A screenshot of the Desmos calculator interface. On the left, there is a blue vertical bar with a small '1' at the top. To its right, the expression '45% of 600' is entered. Further right, there is a grey 'X' icon. Below the expression, the result '= 270' is displayed in a grey box.

Percent Change (Desmos)

To increase or decrease a number by a particular percentage, write the original number, then plus or minus the percentage value.

Example: A population of 1200 mice grows by 15% after one year. Find the new population of mice:



A screenshot of the Desmos calculator interface. On the left, there is a blue vertical bar with a small '1' at the top. To its right, the expression '1200 + 15% of 1200' is entered. Further right, there is a grey 'X' icon. Below the expression, the result '= 1380' is displayed in a grey box.

Note that the starting number (1200) appears twice in the expression, because we are adding a fraction of a number to *the same number*

Solving Equations Involving Percentages (Desmos)

To solve equations that involve percentages or percent change, utilize the same literal expressions as above, including any unknown values as variables, and create regression statements with the $\sim$ symbol.

Example: A jacket is discounted by 20% to a price of \$72. What is the original price of the jacket?

1

$P - 20\% \text{ of } P \sim 72$

×

REGRESSION PARAMETERS

$P = 90$

STATISTICS

RESIDUALS

$RMSE = 0$
 e_1

Note that we chose the letter P for the original price, because that letter represents or reflects the meaning of the variable. It's important to choose variables that connect to the values they represent, to avoid mixing up variable names. Additionally, remember that you are not allowed to use x or y as variable names in regression statements.

Math Skill 5: Surveys and Statistical Reasoning

One of the new topics covered by the SAT Math, and one that the majority of students have not studied in their own math classes, is statistical reasoning. These questions deal with the correct way to gather statistical information--how to conduct a survey, how to select participants, how to interpret the results, etc. Here are some basic guidelines to follow:

A Good Survey....

- Selects participants randomly
- Doesn't require participants to perform a positive action (mail a postcard, call a number, etc)
- Potentially includes every member of the community being measured

Assuming the survey is done correctly, the results...

- Can only indicate features of the community being measured
- Is only a likelihood, rather than a known fact
- Cannot be generalized to indicate any other feature, no matter how similar

Margin of Error

When you take a sample of a larger population, there is always some uncertainty about how well your sample really matches the larger group. This uncertainty is measured as "margin of error," and it is typically a percent value by which your sample measurement might plausibly vary from the actual population measurement. Some tips:

- Start with the sample measure, then subtract the margin of error; this is the lowest plausible value. Now add the margin of error; this is the highest plausible value.
- The margin of error applies to the overall population measurement, not the sample measurement. In our example, we KNOW that 35% of our sample does not believe in evolution. The error comes into the picture when we try to predict what percent of EVERYONE in Florida does not believe in evolution.
- Even with the margin of error factored in, we can only say that certain values are **plausible**. This is because every survey might be catastrophically inaccurate, flawed, etc.

- **The only way to reduce the margin of error is to increase the sample size.** The bigger the sample size, the more accurate your survey becomes. If your sample includes every person in Florida, then.... Voila! No error!

Example: Let's say you sample 100 Floridians and find that 35% of them do not believe in evolution, with a margin of error of 6%. That means that, among all Floridians, it is plausible that between 29% and 41% of them do not believe in evolution.

Typically NOT important:

- Sample size: Even a very small number is enough, assuming the survey is done correctly
- People who choose not to participate: With a random sample, it's not a problem if some people choose not to answer questions.

Math Skill 6: Mean, Median, Mode, Standard Deviation

While the SAT does not cover any advanced statistics topics, the test does expect you to know the three major **measures of central tendency** (mean, median, and mode) as well as the most common **measure of variation** (standard deviation). Here are the basic definitions:

Mean:

The average of a set of numbers. Add up the elements in the set and divide by the number of elements, or $\text{Average} = \text{Sum} / \#$

Median:

The value that falls in the middle of an ordered list of numbers. If there are two numbers in the middle, take the average of the two numbers. Note that a list that's not in order needs to be sorted before you can find the median.

Mode:

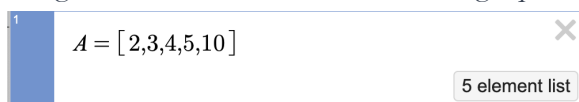
The element that appears the most often in a set. Note that a set can have more than one mode, if there are elements that appear the same number of times.

Standard Deviation:

A measure of the average amount that numbers in a list vary from the mean. In a list that's very spread out, such as [1, 20, 30, 60], the standard deviation is higher. In a list where most numbers are about the same value, such as [10, 10, 11, 12], the standard deviation is lower. Note that you do not need to know how to calculate standard deviation, only what it means.

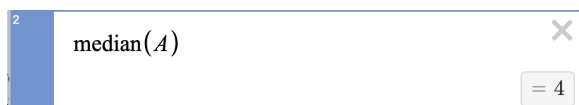
Hint: you can find the mean, median, and standard deviation of a list by using Desmos:

- 1) Assign the list to a variable name, using square brackets:



1 5 element list

- 2) Type in the operation you want, or look for the operation under the Functions menu (look for the Statistics category in the Function menu):



2 = 4

Math Skill 7: Probability

The probability that an event will happen is usually expressed as a fraction representing a value from 0 to 1, where 0 means the event can never happen and 1 means the event will always happen. To find the probability, use this basic formula:

$$\text{Probability} = \frac{\text{Favorable Outcomes}}{\text{Total Outcomes}}$$

The term “favorable outcomes” refers to those outcomes that you are looking for, while “total outcomes” refers to the number of ways the situation can resolve in general.

Example: You roll a single die numbered from 1 to 6. What is the probability that you roll a multiple of 3?

Answer: There are 6 total possible outcomes (1 through 6). Out of those there are 2 outcomes that fit the description (3 and 6 are both multiples of 3). Therefore, the probability of rolling a multiple of 3 = $(2/6)$ or $(1/3)$.

Note: The word “given” in a probability question means the *total* number is restricted to only what is given.

Example: You roll a single die numbered 1-12. What is the probability that you roll a multiple of 3, given that the number you roll is larger than 8?

Answer: Because we’re “given” that the number is larger than 8, only the numbers 9, 10, 11, and 12 are possible at all. Out of those, how many are multiples of 3? Only 9 and 12 fit the description. so the answer is $2/4$ or 0.5

Math Skill 8: Exponents

Exponent rules, also known as laws of exponents, are mathematical principles that govern the manipulation and simplification of expressions involving exponents. Exponents represent the number of times a base is multiplied by itself. Here are the basic exponent rules:

Product Rule:

When you multiply two powers with the same base, you add the exponents: $x^a \cdot x^b = x^{a+b}$

Quotient Rule:

When you divide two powers with the same base, you subtract the exponent of the denominator from the exponent of the numerator: $\frac{x^a}{x^b} = x^{a-b}$

Power Rule:

When you raise a power to another exponent, you multiply the exponents: $(x^a)^b = x^{a \cdot b}$

Zero Exponent Rule:

Any nonzero number raised to the power of 0 is equal to 1: $a^0 = 1, a \neq 0$

Negative Exponent Rule:

A negative exponent is equivalent to the reciprocal of the corresponding positive exponent:

$$x^{-a} = \frac{1}{x^a}$$

Product of Powers Rule:

When you have a product of several powers with the same exponent, you can combine them into a single power by multiplying the bases: $x^a \cdot y^a \cdot z^a = (xyz)^a$

Quotient of Powers Rule:

When you have a quotient of powers with the same exponent, you can combine them into a single power by dividing the bases: $\frac{x^a}{y^a} = \left(\frac{x}{y}\right)^a$

Math Skill 9: Factoring and Roots

Algebraic factoring is a technique used in algebra to express a mathematical expression as the product of simpler expressions or factors. The goal of factoring is to break down a polynomial or algebraic expression into its constituent parts, making it easier to understand, manipulate, and solve.

Factoring

1. Common Factor Factoring:

Identify and factor out the greatest common factor (GCF) of the terms in the expression. This involves finding the largest factor that divides evenly into each term.

Example: For the expression $6x + 9$, the GCF is 3, so you can factor it as $3(2x + 3)$.

2. Factoring Quadratic Expressions:

Quadratic expressions of the form $x^2 + bx + c$, where b and c are constant, can be factored into the product of two binomials $(x + p)$ and $(x + q)$, where:

$$p + q = b$$

$$p \cdot q = c.$$

Example: $x^2 - 4x + 8 = (x - 6)(x + 2)$

3. Difference of Squares

Expressions in the form $a^2 - b^2$ can be factored into the product of two binomials, $(a - b)$ and $(a + b)$

Example: $4x^2 - 9y^2 = (2x + 3y)(2x - 3y)$

4. Perfect Square Trinomials

Expressions in the form $a^2 + 2ab + b^2$ can be factored into the product of two binomials, $(a + b)$ and $(a + b)$, or $(a + b)^2$

Example: $9p^2 - 6pq + q^2 = (3p - q)^2$

Factoring is an essential skill in SAT algebra, particularly when trying to solve quadratic equations.

Roots

The **roots** of an equation, also known as the **solutions** or **zeros**, are the values of the variable in the equation that make the equation equal to zero.

To solve for the roots of a polynomial equation, including a quadratic equation, you must (1) ensure that the equation is equal to zero, (2) write the equation in its most factored form, then (3) use the Zero Product Property:

Zero Product Property: If the product of two or more factors is equal to zero, then at least one of the factors must be zero. In other words, if $ab = 0$, then either $a = 0$ or $b = 0$ (or both).

For a polynomial that is equal zero and is written in fully factored form, you can set each factor equal to zero separately. The solution for each factor's equation is a solution to the equation as a whole.

For example, to solve $x^2 - 5x + 6 = 0$

1. Factor the left side: $(x - 3)(x - 2) = 0$
2. Set each factor equal to zero: $x - 3 = 0$, $x - 2 = 0$
3. Solve each factor's individual equation: $x = 3$, $x = 2$

Final Note (Important!): Because each factor in an equation produces its own root or solution, you can think of factors and roots as being linked to each other. In other words, knowing the factors automatically lets you know what the roots are, and knowing the roots allows you to know what the factors are. If the SAT gives you factors (or something you can re-write into factors), then you probably need to turn them into roots. If the SAT gives you roots, then you probably need to turn them into factors.

Math Skill 10: Functions

Algebraic factoring is a technique used in algebra to express a mathematical expression as the product of simpler In mathematics, a function is a relationship or correspondence between two sets of numbers, known as the **domain** and the **range**. A function assigns each element from the domain to exactly one element in the **range**. The notation $f(x)$ is commonly used to represent a function, where x is an element in the **domain** and $f(x)$ is an element in the **range**. You can think of the domain as the “input” and the domain as the “output.”

Here are some key concepts:

Domain: The set of all possible input values for which the function is defined.

Range: The set of all possible output values that the function can produce. When you graph a function, the values of $f(x)$ are assigned to the variable y .

Rule or Mapping: The rule or mapping describes how each element in the domain is paired with a unique element in the range. This rule defines the relationship between the input and output values.

Function Notation: Functions are often denoted using function notation, such as $f(x)$ or $h(t)$, where the variable inside the parentheses represents the input.

Example: Consider the function $f(x) = 2x + 1$

This function takes an input x , multiplies it by 2, and then adds 1 to produce the output $f(x)$. The domain and range are {all real numbers} and the **rule** is $2x+1$:

When we write something like $f(3)$, it means that $x=3$ is the input to the function, so we should plug 3 in for every x in the equation: $f(x) = 2 \cdot 3 + 1 = 7$

Alternatively, we might have a question that asks, “For what value of x does $f(x) = 5$?” In that case, 5 is the **output** we need, and we’re missing the input. So:

$$2x + 1 = 5 \rightarrow x = 2$$

On the SAT, function questions will almost always provide the rule (that is, the equation), and solving them almost always involves plugging in the provided input value into the variable. In some

cases, the rule might be a table or a graph instead of an equation. The same basic concepts apply, regardless of how the rule is presented.

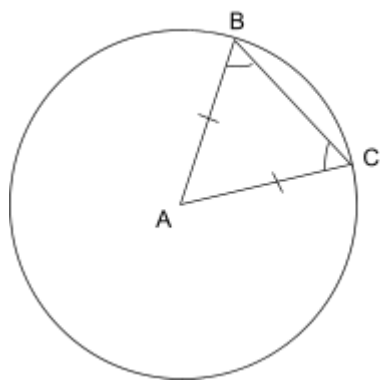
Math Skill 11: Circles

In the Digital SAT Math section, you will encounter questions that pertain to circles, either as geometric shapes or as equations on a graph. Here are some basic rules and concepts that you need to know.

Circles, Angles, and Arcs

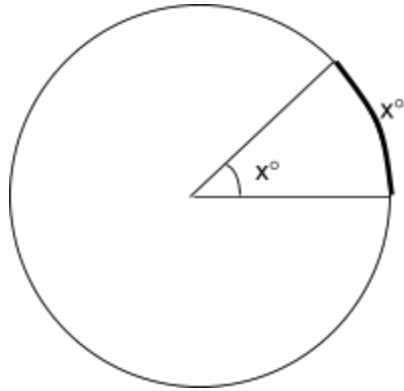
A circle is defined as the set of points that are **equidistant** from a center point. The distance between the center of a circle and any point on the circle is called the **radius**. A line segment drawn from one point on the circle to another point on the circle, through the center, is called the **diameter**. Some relationships and formulae:

- The length of the diameter is twice the length of the radius ($d = 2r$).
- The length of a radius is equal to the length of any other radius drawn in the same circle.
- A triangle with one vertex on the center of the circle and two of its vertices on the edge of a circle is always **isosceles**:

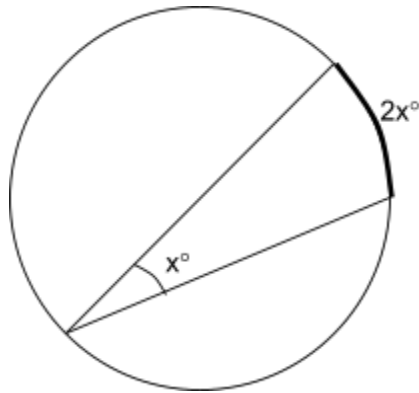


In this diagram, triangle ABC is isosceles, with $AB = AC$ and $\angle ABC = \angle ACB$.

- An **arc** is a length along the circumference of a circle. It can be measured in angle units (degrees, radians, etc) or in linear units.
 - An arc created by a central angle has an **arc measure** (in degrees) equal to the measure of the central angle that created it:



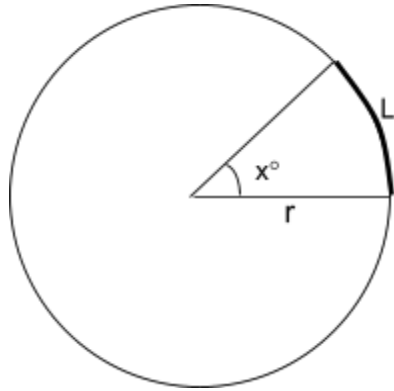
- An arc created by an inscribed angle has an **arc measure** that is equal to **double** the measure (in degrees) of the angle that created it:



- An arc that coincides with the entire outer perimeter of the circle is called the **circumference**. Its length is proportional to the length of the circle's radius:

$$C = 2\pi r$$

- The **arc measure** of a circle's circumference is 360° .
- The **arc length** of an arc is proportional to its **arc measure**, or to the central angle that creates it:

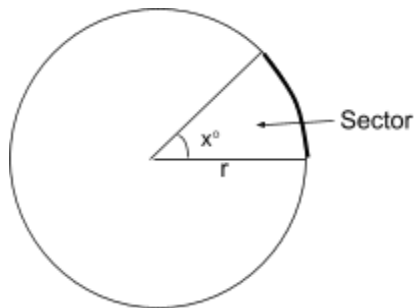


$$\frac{L}{x} = \frac{2\pi r}{360}$$

- The **area** of a circle is proportional to the square of the radius:

$$A = \pi r^2$$

- A central angle in a circle creates a subsection of a circle's area called a **sector**. The area of a sector is proportional to central angle that creates it:



$$\frac{A_{\text{sector}}}{x} = \frac{\pi r^2}{360}$$

Math Skill 12: Lines and Angles

A significant percent of geometry questions on the DSAT involve triangles and angles, as well as several key features of lines. Here are some basic rules and formulas you need to know.

Lines

A line is an undefined term in geometry, but you can probably imagine what it looks like.

- A line can be drawn between any two points. The **slope** of the line represents the angle or slant that the line travels relative to the horizontal. It's usually represented by the variable **m**. If you know the coordinates of the two points, you can find the slope of the line:

$$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1}$$

- The **x-intercept** of a line is the point where the line crosses the x-axis. You can find the x-intercept by substituting 0 for the y value in the line (any point on the x-axis has a y coordinate of 0).
- The **y-intercept** of a line is the point where the line crosses the y-axis. You can find the y-intercept by substituting 0 for the x value in the line (any point on the y-axis has an x coordinate of 0).
- Two parallel lines have **equal slopes**.
- Two perpendicular lines have **opposite reciprocal slopes**. For example: $\frac{2}{5} \leftrightarrow -\frac{5}{2}$

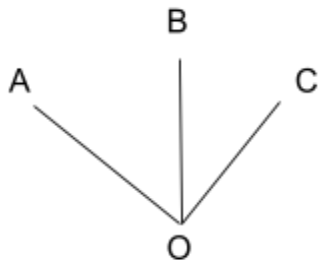
Angles

The intersection of two lines or line segments creates one or more **angles**.

- Angles are typically measured in degrees or radians, and the DSAT often asks students to convert from one to the other. The conversion scale is as follows: π radians = 180°
- To convert from angles to radians, or vice versa, use **dimensional analysis**. For example, to convert $\frac{2\pi}{3}$ radians to degrees:

$$\frac{2\pi}{3} \cdot \frac{180^\circ}{\pi} = \frac{360^\circ \cdot \pi}{3\pi} = \frac{360^\circ}{3} = 120^\circ$$

- When two parallel lines are crossed by a **transversal**, they form 8 total angles. Four of the angles are acute (small) and four of the angles are obtuse (bigger than 90°). **Any small angle is equal to any bigger angle.** The **sum of a small angle and a big angle is always 180°**
- Two angles that share a vertex and one side are called **adjacent** angles:



Angle AOB and angle COB are adjacent.

- Two adjacent angles that form a straight line are called a **linear pair**. The sum of their measures is 180° :



$$\angle AOC + \angle AOB = 180^\circ$$

Math Skill 13: Equivalent Expressions

A great number of SAT equations explore the concept of **equivalence**. You can identify these questions because they use certain keywords: “equivalent,” “can be written as,” “equation is true for all values of x ,” “equation has infinitely many solutions.”

Expressions that are **equivalent** are **identical**, meaning they always produce the same answers from equal inputs. In most cases, one expression is the result of simplifying, factoring, expanding, etc, the other expression.

Example:

Which expression is equivalent to $\frac{4}{4x-5} - \frac{1}{x+1}$?

- A. $\frac{1}{(x+1)(4x-5)}$
- B. $\frac{3}{3x-6}$
- C. $-\frac{1}{(x+1)(4x-5)}$
- D. $\frac{9}{(x+1)(4x-5)}$

The most powerful strategies for solving these questions use the element of equivalence in one way or another. It’s important to familiarize yourself with all the strategies, because different questions will respond more readily to some strategies than to others.

Strategy 1 (Algebra):

Using algebraic steps that you know are allowed, take one expression and reduce or simplify it until it looks like the other expression. For example, if the original expression is two fractions being added, try getting a common denominator and adding them normally. You should get something similar to the desired answer.

Strategy 2 (Desmos Graphing)

For multiple choice questions where each answer is an expression, and all the expressions can be graphed.

When two expressions are equivalent, they have the **same graph**. If the initial expression contains one variable and can be graphed, try graphing the expression in Desmos. Use a slider for any constants in the expression (this may not work if there are too many constants. Then, graph the

expression you are trying to match or graph the individual answer choices (depending on the question). If you get the constants right, the two graphs will be directly on top of each other.

Strategy 3 (Desmos Evaluating)

For multiple choice questions where each answer is an expression, but there are too many variables to graph the expressions.

When two expressions are equivalent, they always produce the same output for the same input. Write out the original expression on a Desmos tile, adding sliders for all variables. Move the sliders randomly to create a random answer. Then, write each answer choice into Desmos, leaving the sliders untouched. The right answer will produce the exact same output that you got for the original expression.

Strategy 4 (Regression)

For questions where the answers are not expressions, but require you to find “constants” in the question

Some equivalent expression questions don’t offer you alternative expressions in the answer choices. Instead, they ask you to find one or more constants that *make* one expression equivalent to another. You can try to use graphing or sliders for these questions, but it can be difficult to pinpoint the exact right value for the constants. Sometimes the constants are very small decimal, or there are multiple constants. In that case, try regression:

- 1) Set the expressions equal to each other
- 2) Follow the steps for inputting a regression equation (See above) (Well, here they are again in case you don’t want to go back)
 - a) Replace any x in the equation with x_1 ; replace any y with y_1
 - b) Replace the equal sign with the regression symbol ($\sim$). You can find that symbol on the Desmos keyboard, or just left of the number 1 on your physical keyboard.
- 3) Add some values for Desmos to plug into the variable (not the constant):
 - a) Use list notation. Ex: $x_1=[2,3,4,5,6]$
 - b) Avoid any values of x or y that would cause division by zero in any part of the equation. If you do this accidentally, Desmos will give you an error message along the lines of “We couldn’t find any region where this model is defined.” Erase the number causing the problem from the list.
- 4) Look back at the regression tile: the “parameters” should have changed to new values. These are the correct values for the constants. Voila!

Many equivalence questions can be solved by more than one of these strategies, with a little adjustment on your part.

Math Skill 14: Finding the Model

Some SAT word problems are based around real-world situations, or describe the behavior or output of a particular function, then ask you to find the equation that matches that situation or function.

For most word problems that ask you for an equation, it's easier to find the right equation by thinking about real values, rather than using the variables to come up with the equation yourself.

Example:

Jennifer bought a box of Crunchy Grain cereal. The nutrition facts on the box state that a serving size of the cereal is $\frac{3}{4}$ cup and provides 210 calories, 50 of which are calories from fat. In addition, each serving of the cereal provides 180 milligrams of potassium, which is 5% of the daily allowance for adults. If p percent of an adult's daily allowance of potassium is provided by x servings of Crunchy Grain cereal per day, which of the following expresses p in terms of x ?

- A. $p = 0.5x$
- B. $p = 5x$
- C. $p = (0.05)^x$
- D. $p = (1.05)^x$

Strategy:

When the SAT asks you to find an equation for a real-world situation, there is a strong possibility that the process contains some kind of trap, especially for the harder questions. Instead of dealing with the algebra directly, it's usually better to think of numbers that work, based on the description in the question. Then see which equation matches your numbers.

In the example question above, you're bombarded with information, most of which is not relevant to the question. However, can we think of a specific number of servings, for which we would know what specific percent of your daily allowance that serving provides?

If you dig through all the noise, you'll see that one serving (which is $\frac{3}{4}$ of a cup, but that is just more extra information!) of this cereal provides 5% of your daily allowance. So, when $x = 1$, $p = 5$. Which answer matches those numbers? Only answer (B) gives you $p=5$ when $x=1$. That's the right one.

You can apply this strategy to any word problem or question where you know something about the input and output of a relation, and your task is to find an equation that matches. It could be about the cost of hiring a plumber, the number of seats in an auditorium, the pattern of numbers in a sequence, etc. The details are not important, only the common element: you know the numbers, and you have to find the equation.

Math Skill 15: Dimensional Analysis

SAT word problems that involve units are often designed to test your ability to convert from one unit to another. For these problems, you can apply a tool from the field of science: **dimensional analysis**.

A unit of measurement can be treated just like any other variable or number: it can be multiplied with itself to make a square unit, or divided by itself to cancel out. **Dimensional Analysis** is the process of using the units to determine the correct form of an equation or expression. You can use dimensional analysis (D.A.) to decide whether a particular equation is correct, or you can use D.A. to decide the correct way to convert a unit or rate.

Example:

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by \$5. Assuming gasoline costs \$4 per gallon, which equation can Alan use to determine how many fewer average miles, m , he should drive each week?

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

Strategy:

Use dimensional analysis! Let's look through the answer choices to see which equations make sense and which equations are impossible for the units. First, let's decide what the numbers in the answers probably represent, in units:

25: this number is given in the question as miles per gallon, which can be written as **miles/gallon**

4: this is probably the \$4 per gallon, so it's measured in **\$/gal**

m: this variable represents the number of miles, so the unit is **miles**

95: ?? This number seems to be (100 miles) - (\$5), which is impossible. We can't subtract dollars from miles. Both answers that have this number already don't make sense

5: Probably the money Alan wants to save, which should be measured in \$

Now let's look at the answer choices one by one:

A) $\frac{\text{miles/gallon}}{\$/\text{gallon}} \cdot \text{miles} = ???$ We decided the number 95 was probably nonsense, so eliminate this answer without going further.

B) $\frac{\text{miles/gallon}}{\$/\text{gallon}} \cdot \text{miles} = \$$ This one makes more sense, so let's continue and reduce:

$$\frac{\text{miles}}{\text{gallon}} \cdot \frac{\text{gallon}}{\$} \cdot \text{miles} = \$ \quad (\text{To divide by a fraction, flip the fraction and multiply})$$

$$\frac{\text{miles}}{1} \cdot \frac{1}{\$} \cdot \frac{\text{miles}}{1} = \$ \quad (\text{Cancel the gallons, add a denominator to miles to multiply})$$

$$\frac{\text{miles}^2}{\$} = \$ \quad (\text{Multiply})$$

This final statement is false, because square miles divided by dollars does not equal dollars. Eliminate this answer.

C) There's that 95 again, so this one is probably wrong. Let's not spend time here.

D) $\frac{\$/\text{gallon}}{\text{miles/gallon}} \cdot \text{miles} = \$$ Let's continue and reduce:

$$\frac{\$}{\text{gallon}} \cdot \frac{\text{gallon}}{\text{miles}} \cdot \text{miles} = \$ \quad (\text{To divide by a fraction, flip the fraction and multiply})$$

$$\frac{\$}{1} \cdot \frac{1}{\text{miles}} \cdot \frac{\text{miles}}{1} = \$ \quad (\text{Cancel the gallons, add a denominator to miles to multiply})$$

$$\frac{\$}{1} \cdot \frac{1}{1} \cdot \frac{1}{1} = \$ \quad (\text{Cancel the miles})$$

$$\$ = \$ \quad (\text{Multiply})$$

This answer works for the units in the question.

Out of all the answer choices, only D actually works for the units provided in the question. Without necessarily understanding why each variable or number is placed where it is, we can select this answer with very high confidence.

Math Skill 16: Systems of Equations Special Cases

Earlier, we covered the process of solving systems of equations—use Desmos to graph the system and look for the point(s) of intersection. However, some SAT questions involve special cases in systems of equations—systems that either have “**no solution**” or “**infinitely many solutions**.”

- A system of equations in which the two equations are **identical (equivalent)** will have infinitely many solutions.
- A system where the two equations create **parallel lines** will have no solutions.

Example:

$$\begin{aligned}\frac{3}{2}y - \frac{1}{4}x &= \frac{2}{3} - \frac{3}{2}y \\ \frac{1}{2}x + \frac{3}{2} &= py + \frac{9}{2}\end{aligned}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

Strategy 1 (Desmos Graphing):

If the question includes multiple choice answers as possible values for the constant, or if the question has relatively simple numbers (no weird fractions or very small numbers), then you might be able to solve through graphing. Graph each equation on a separate line, adding a slider for the constant(s). Then use the sliders to find the constant value that either creates parallel lines (“no solution”) or two graphs superimposed on each other (“infinitely many solutions”). For multiple choice questions, jump the constant to the values in the answers, to save time.

Strategy 2 (Algebra)

For questions that involve unusual fractions or small numbers, but do not have multiple choice answers, you might do better with direct algebra. In the case of “no solution” for a system of linear equations, find the slope of each line (by putting them into $y=mx+b$ form), then set the slopes equal to each other. To have “no solution,” the lines have to be parallel, which means their slopes have to be identical. In the case of “infinitely many solutions,” follow the steps above, but **everything** has to match in the two equations, not just the slope.

Strategy 3 (Ratios)

When two linear equations are either identical or parallel, it's because the coefficients of the x- and y-terms in one equation are proportional to the coefficients of the x- and y-terms in the other. In other words, one equation is just a multiple of the other. We can use this relationship to our advantage by creating a proportion with the coefficients and solving for the missing constants.

Take this example:

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$$48x - 64y = 48y + 24$$

$$ry = \frac{1}{8} - 12x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

If we combine like terms on each side, we get something like this:

$$48x - 112y = 24$$

$$12x + ry = \frac{1}{8}$$

Use a proportion:

$$\frac{48}{-112} = \frac{12}{r}$$

Solve for r :

$$r = -28$$

The proportion strategy works best for questions where the answers are fractions, or where the system has two or more missing constants and you have to find the value of an expression involving those constants. However, the strategy works well for all types of special systems, so it's worth learning.